HIRDIANSHU SINGLA

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EDUCATION

BASc, Computer Engineering — University of British Columbia, Vancouver, BC Expected: 12/2027
Relevant Coursework: Machine Learning & Data Mining (regression, classification, clustering, neural networks, gradient descent) · Algorithm Design & Analysis · Intelligent Systems (search, CSPs, planning, probabilistic reasoning)

TECHNICAL PROJECTS

Chess AI Platform · Python · PyTorch · TypeScript · Socket.io Jan – Apr 2026

- Conducted **supervised learning experiments** training a feedforward neural network on 50,000 Stockfish-evaluated positions with **70-dimensional engineered board feature vectors**; systematically tuned hyperparameters (learning rate, dropout, batch size) and analyzed convergence via R^2 and loss curves to identify underfitting/overfitting regimes
- Designed a **controlled evaluation study** comparing learned neural evaluation against classical minimax heuristics across 500+ test positions; analyzed **accuracy vs. search depth tradeoffs** to quantify where ML evaluation outperforms hand-crafted scoring — integrated both as interchangeable evaluators in a shared alpha-beta engine
- Deployed **real-time two-player multiplayer** via Socket.io with room-based matchmaking, server-side move validation, and disconnect recovery; full **TypeScript/Express REST API** with persistent game state management

ARC4 Stream Cipher Cracker · SystemVerilog · Cyclone V · Quartus Jan – Apr 2026

- Architected a **128-core parallel ARC4 cryptanalysis engine** on Cyclone V FPGA via **parametric generate instantiation**, achieving **5.74M keys/second** across a 24-bit keyspace (16.7M total keys) — brute-forced to completion in under 3 secs
- Optimised per-core **FSM with early-abort PRGA** and inline ASCII validation; reduced cycles-per-wrong-key from ~630 → ~14 (**~45× speedup**); cut device utilisation 97% → 65% by eliminating 128 private PT on-chip memory blocks
- Synthesized **ALTPLL IP** to boost clock 50 → 70 MHz; implemented custom **MBOX mailbox protocol** for automated evaluator I/O; built 32-bit on-chip **cycle counter** for hardware performance profiling and reproducible benchmarking

F1TENTH Autonomous Racing · ROS2 · Python · LiDAR · OpenCV · MPCC Sept – Dec 2025

- Investigated **Model Predictive Contouring Control (MPCC)** for real-time overtaking on a 1/10-scale autonomous car; fused **LiDAR point clouds** and **monocular vision** for simultaneous obstacle detection and path tracking — iteratively tuned cost function weights to balance speed vs. tracking error under real-time latency constraints
- Built **LiDAR obstacle detection and vision processing pipeline** (OpenCV) feeding environment state to the motion planner; applied **TTC-based safety braking** as a fail-safe; structured **telemetry logging** enabled **reproducible experiments** and systematic regression testing of controller parameters across track configurations

AI Financial Assistant & Messenger · Node.js · Llama3 · LLaVA · Whisper Jan – Apr 2026

- Ranked Top 3 out of 37 groups for developing an AI Financial Assistant, recognized for innovation, technical depth, and real-world impact
- Built a real-time **multi-room chat application** (Node.js/Express/MongoDB/WebSockets) with an integrated **multimodal AI panel** supporting text, **voice transcription (Whisper)**, and **chart-image analysis (LLaVA)** — identifying candlestick patterns, trend direction, and support/resistance from uploaded screenshots
- Deployed a **Llama3-powered context-aware ad recommendation sidebar** reading live chat messages and refreshing relevant suggestions every 30 s; **Python Flask AI server** exposing six REST inference endpoints for LLM, vision, and speech models with **Alpha Vantage** live stock price integration

WORK EXPERIENCE

Trainee Analyst — Winux Software Solutions Pvt Ltd, Mohali, Punjab Apr–Aug 2024; Jun–Sep 2025

- Led 8+ **requirements discovery sessions** with ~10 stakeholders; authored 30+ **user stories** and **BPMN process flows**; iteratively refined specifications based on stakeholder feedback, shifting defect discovery one sprint earlier
- Designed and implemented **Python data pipeline** for automated log parsing, **statistical analysis** of pass/fail distributions, and first-fail triage report generation — applied **systematic testing methodology** with QA to derive edge-case scenarios improving coverage

TECHNICAL SKILLS

Languages: Python · Java · C/C++ · JavaScript · TypeScript · SQL · Bash · SystemVerilog · Verilog
ML & Data: PyTorch · scikit-learn · NumPy · pandas · Neural network design & training · Feedforward networks · Backpropagation · Gradient descent · Hyperparameter tuning · Feature engineering · Model evaluation (R^2, loss, accuracy) · Regression · Classification · Clustering · Transfer learning · Reinforcement learning concepts · Data preprocessing · Train/val/test splits
Software & Systems: Git · Linux · GDB · REST APIs · WebSockets · MongoDB · Docker (basic) · Agile/Scrum · Unit testing · Requirements analysis · CI/CD concepts · Node.js · Express · Flask
Robotics & Vision: ROS2 · LiDAR sensing & point cloud processing · Motion planning (MPCC) · OpenCV · Sensor fusion · TTC-based safety systems · Ackermann control · Autonomous vehicle control · Path tracking · Obstacle detection
Hardware & FPGA: Quartus · ModelSim/Questa · Cyclone V · DE1-SoC · Parallel architecture · FSM design · FIFO design · Formal verification (SVAs) · Clock synthesis (ALTPLL) · LFSR · Synthesis & timing analysis · Hardware performance profiling